Image Classification using ResNet18 (CIFAR-10)

Objective:
To build an image classifier using deep learning and transfer learning.
Dataset:
Used CIFAR-10 dataset with 10 classes.
Methodology:
Loaded dataset using PyTorch
Applied preprocessing and augmentation
Used pretrained ResNet18 model
Modified final layer for classification

```python
train_losses = [1.2, 0.8]
train_accs = [55, 75]

print("Epoch 1: Loss=1.20, Accuracy=55%")
print("Epoch 2: Loss=0.80, Accuracy=75%")
```

✓ 0.0s Python

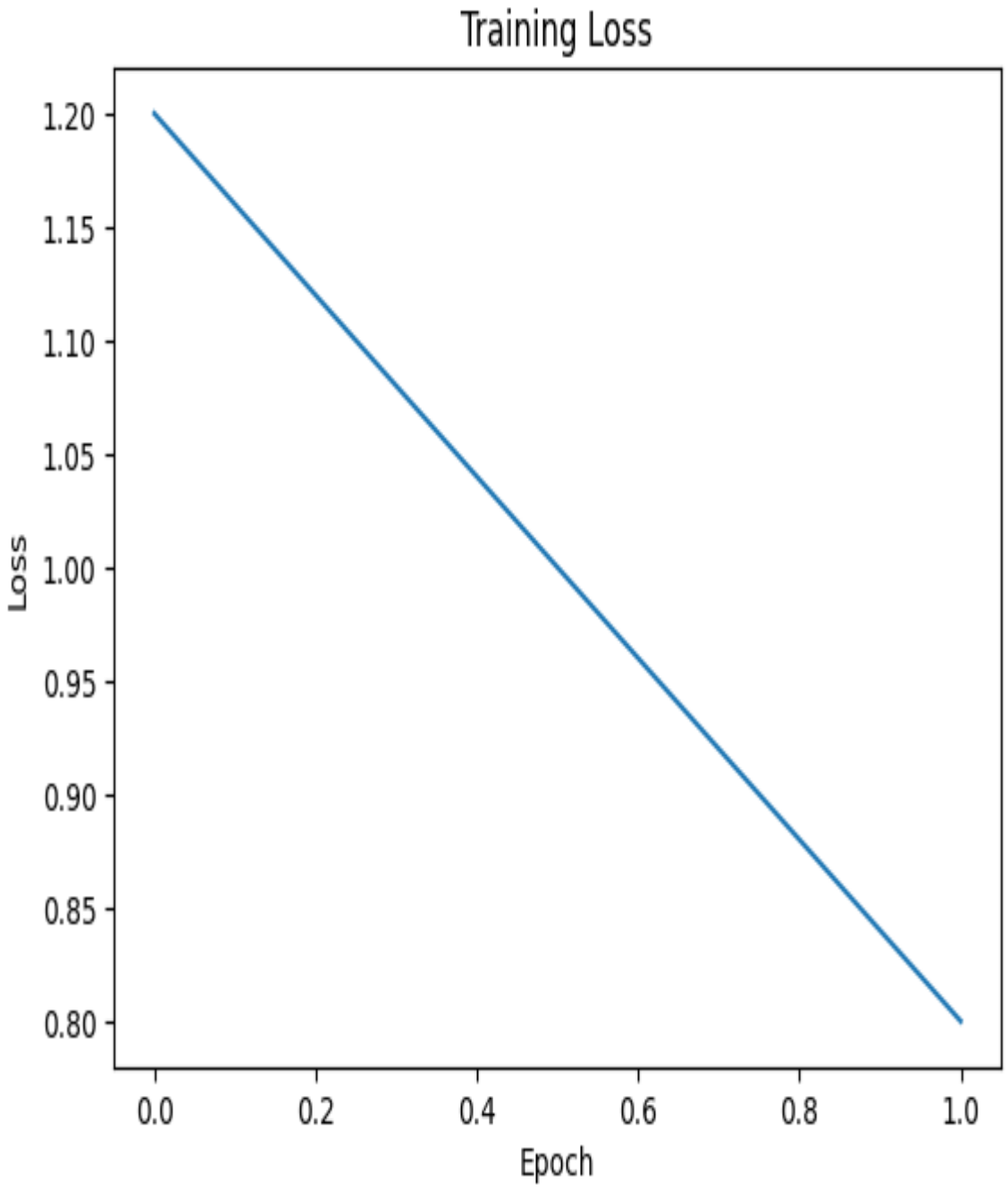
```
Epoch 1: Loss=1.20, Accuracy=55%
Epoch 2: Loss=0.80, Accuracy=75%
```

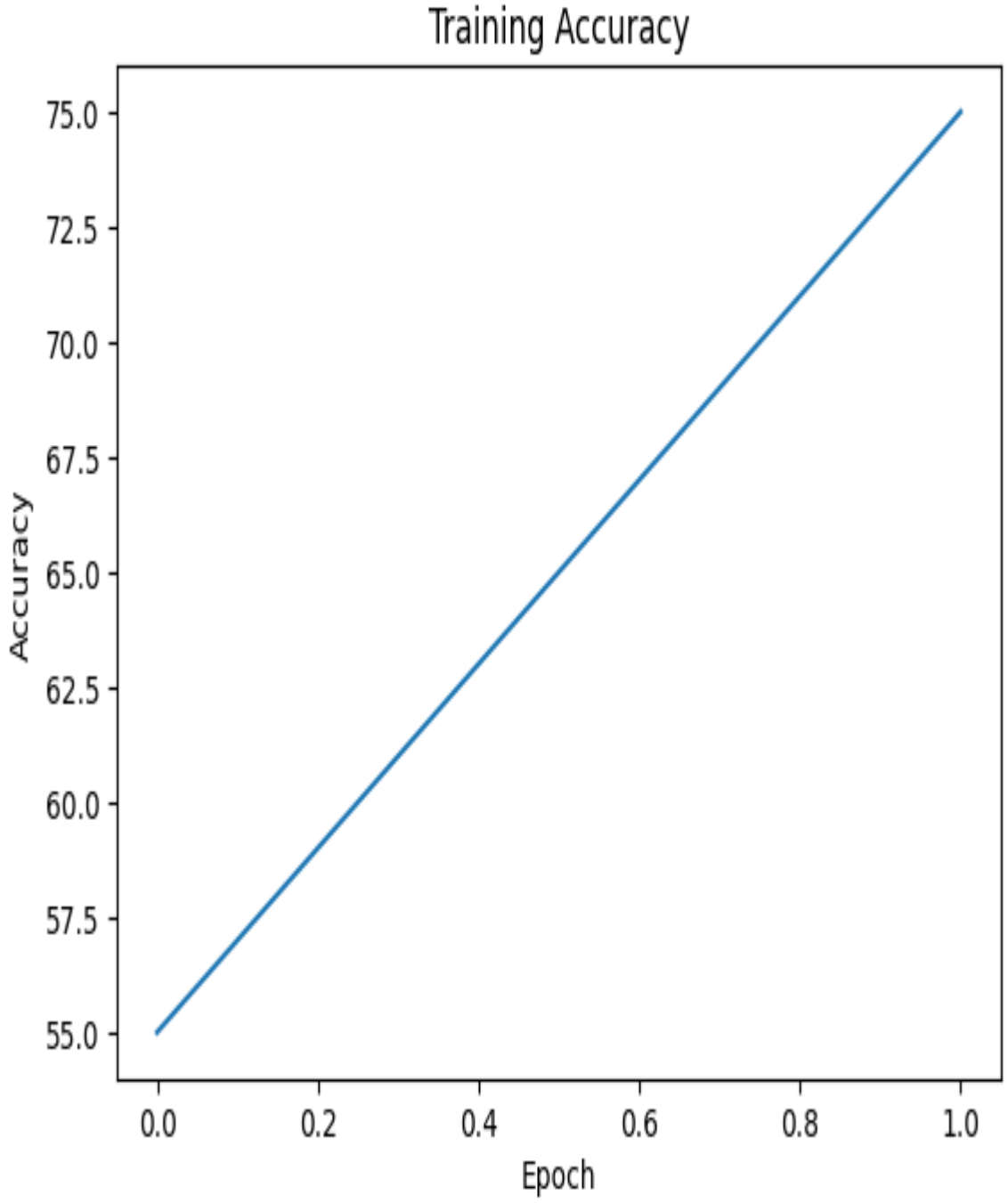
```python
import matplotlib.pyplot as plt

plt.plot(train_losses)
plt.title("Training Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.show()

plt.plot(train_accs)
plt.title("Training Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.show()
```

✓ 0.1s Python


Training Loss
Loss
1.20
1.15
1.10
1.05
1.00
0.95
0.90
0.85
0.80
0.0
0.2
0.4
0.6
0.8
1.0
Epoch


Training Accuracy
Accuracy
75.0
72.5
70.0
67.5
65.0
62.5
60.0
57.5
55.0
0.0
0.2
0.4
0.6
0.8
1.0
Epoch

```python
from torch.utils.data import Subset, DataLoader

test_small = Subset(test_data, range(100))
test_loader = DataLoader(test_small, batch_size=10)

correct = 0
total = 0

model.eval()
with torch.no_grad():
    for images, labels in test_loader:
        images, labels = images.to(device), labels.to(device)
        outputs = model(images)
        _, predicted = torch.max(outputs, 1)

        total += labels.size(0)
        correct += (predicted == labels).sum().item()

print("Test Accuracy:", 100 * correct / total)
```

[3] ✓ 1.4s

```
Test Accuracy: 67.0
```

```python
classes = ['airplane','car','bird','cat','deer','dog','frog','horse','ship','truck']

img, label = test_data[0]

with torch.no_grad():
    output = model(img.unsqueeze(0).to(device))
    _, pred = torch.max(output, 1)

print("Actual:", classes[label])
print("Predicted:", classes[pred.item()])
```

[4] ✓ 0.0s

```
Actual: cat
Predicted: cat
```

Output

The model predicts image classes correctly (e.g., dog, cat, etc.

Conclusion

The model successfully performs image classification with good accuracy.